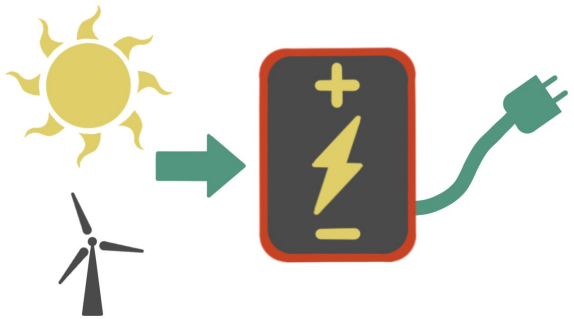
Safety Analysis and Application to Battery Management Systems (BMS)



Ania's BMS Course

by Ania Mitros, PhD

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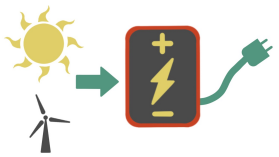
BMS Safety



Video 1
- Car/home battery safety
  - Industry standards & terms
  - Key considerations

Focus on managing large lithium ion batteries

~~Small applications~~

Video 2
- Fault tolerance
  - Fault tolerance levels
  - Types of faults

General safety concepts not specific to batteries, applicable to automotive or similarly safety-conscious systems with battery examples

Video 3
- Analysis tools
  - FTA = Fault Tree Analysis

Video 4
  - DFMEA = Design Failure Mode and Effects Analysis

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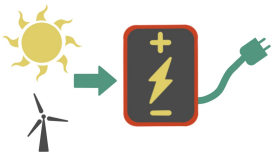
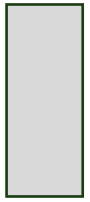
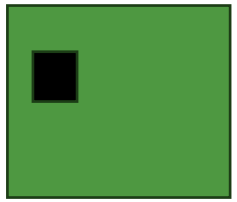
What does “safety” encompass?



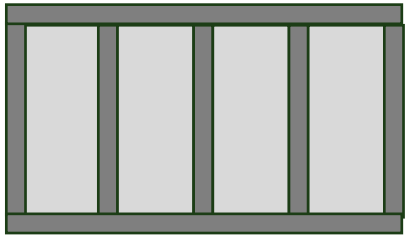


Cells:
- Chemistry (prone to exothermic behavior?)
- Manufacturing quality (defect rate)



Electrical engineering:
- Sensor selection
- Actuator control: Contactors, fuses
- Hardware-only protections

Electro-mechanical design:
- Contactors/switches
- Fuses



Pack (mechanical construction):
- Heating/cooling
- Propagation barriers
- Venting

Application:
- How impactful is a fire?
- How impactful is a system shutdown?

Team organization

Software:
- Safety-related decisions (allow operation, stop charge, etc)
- Safety diagnostics

Testing:
- Does implementation achieve intent?

Manufacturing:
- Does defect rate support design (ex. Choices made in DFMEA)?

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Automotive safety industry terms

Functional Safety: The automated part of a safety system.

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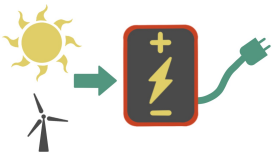

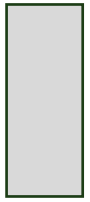

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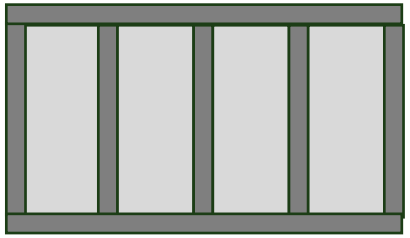

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Functional Safety

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- **ISO 26262** – Automotive ← FTA and FMEA are among the tools necessary to comply with ISO 26262
- EN 50128 – Railway
- IEC 62304 – Medical Device
- IEC 62061 – Manufacturing
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More about standards: https://www.perforce.com/resources/qac/what-is-functional-safety

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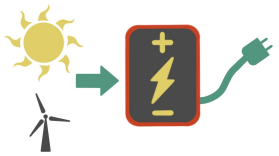

ad hoc approach can be broader

Cells:
- Intensive or dense (exothermic behavior?)
- Manufacturability (defect rate)

Good for vertically-integrated car-makers which make their own parts

Pack (mechanical construction):
- Housing/cooling
- Propagation barriers
- Venting

Electrical engineering:
- Sensor selection
- Actuator control: Contactors, fuses
- Hardware only protections

Electro-mechanical design:
- Contactors/switches
- Fuses

ISO26262 covers Functional Safety

Good for subcomponents built by one company for another

Application:
- How impactful is a fire?
- How impactful is a system shutdown?

Team organization

Software:
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ISO26262

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ASIL: Automotive Safety Integrity Level: A risk classification system defined by the ISO 26262 standard for the functional safety of road vehicles.

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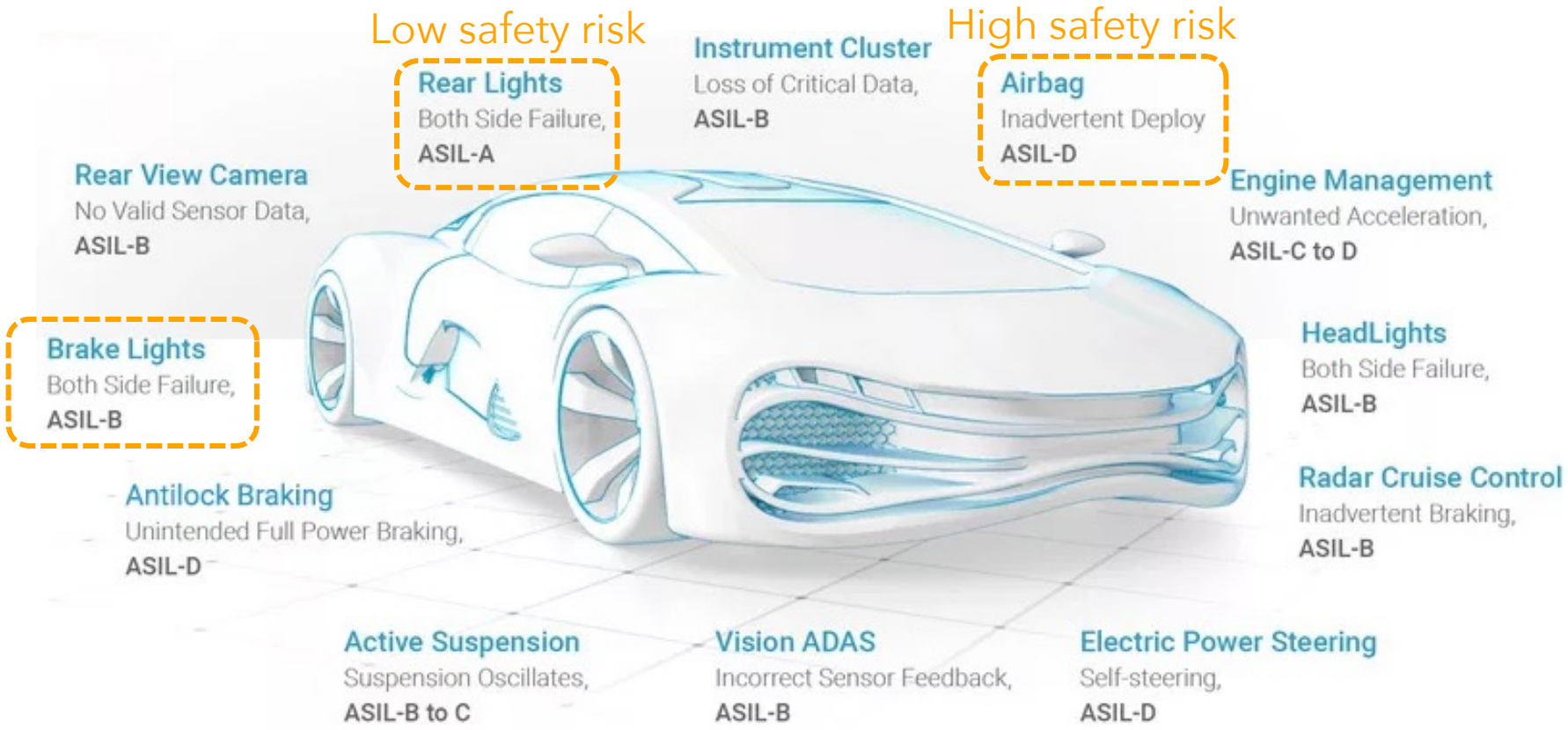
ASILs (Automotive Safety Integrity Levels)



From: https://www.synopsys.com/automotive/what-is-asil.html

Battery electronics: ASIL-D

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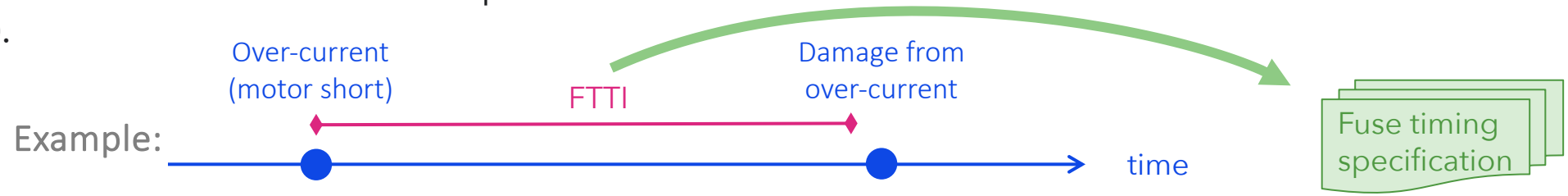
FTTI: Fault Tolerant Time Interval: The time period from the occurrence of a fault to the violation of a Safety Goal (SG).



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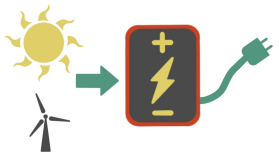
UL1973: Stationary Battery Safety Standard



Includes:

- **Construction requirements:** Materials, enclosures, electrical spacing, insulation, grounding, thermal management
- **System safety analyses**
- **Safety performance tests:** Overcharge, short circuit, over-discharge protection, temperature and operating limits check test, imbalanced charging test, dielectric voltage test, continuity test, thermal system test
- **Production tests**
- **Mechanical tests:** Vibration, shock, crush,
- **Environmental tests:** Thermal cycling, resistance to moisture, salt fog
- **Fire exposure tests:** External & Internal. Internal test requires initiating thermal runaway in one cell and observing for 1 hour, requiring no propagation of fire and no explosion.

Thorough and readable description: https://www.everexceed.com/overview-of-lithium-battery-safety-testing-ul-1973_n433

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Safety: Car vs home battery

Should stationary battery or automotive battery standard require more rigor around safety?

Car safety:

Driving

Injury

Potential for accident, stranding

Driver is awake and aware

Parked in garage

Loss of power: Likely low repercussions

Potential for fire to spread to structure

Home occupants may be asleep

Big issues:
- Fire
- Loss of power

similar

Stationary battery safety:

Potential for fire to spread to structure

Low risk due to loss of power

Home occupants may be asleep

Higher risk if battery is sole energy source (off-grid) or powers safety-critical equipment

Power off is <u>not</u> a safe state in some systems

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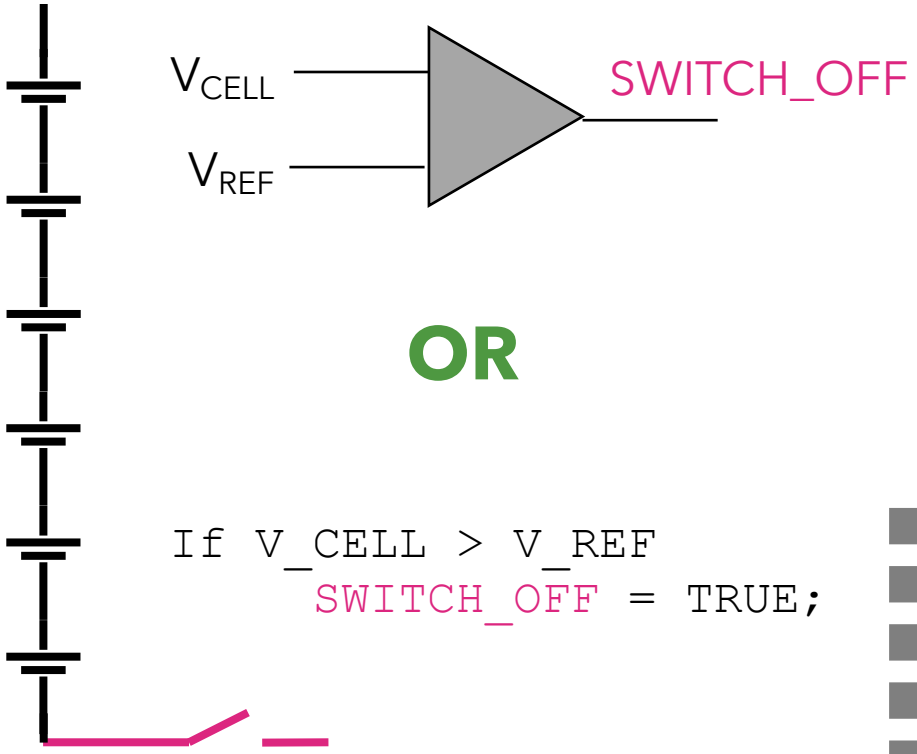
Hardware- vs software-mediated responses

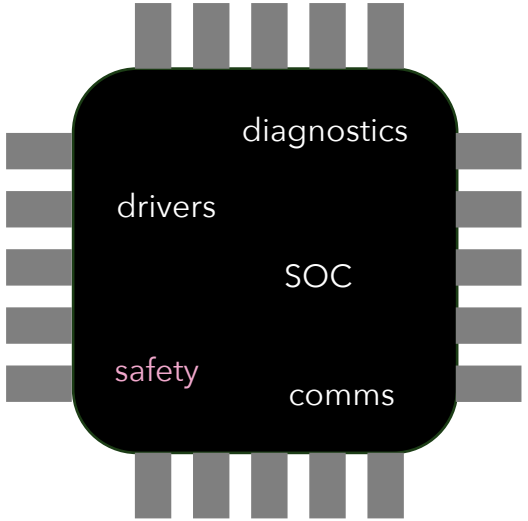
Any change → validate per standard

System limits and SOC estimation are appealing to finalize late in design, when system is nearly complete.

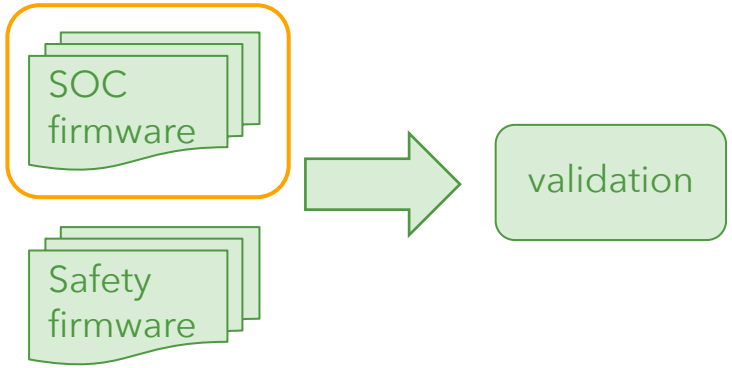


OR

```
If V_CELL > V_REF
    SWITCH_OFF = TRUE;
```



Schedule dependency



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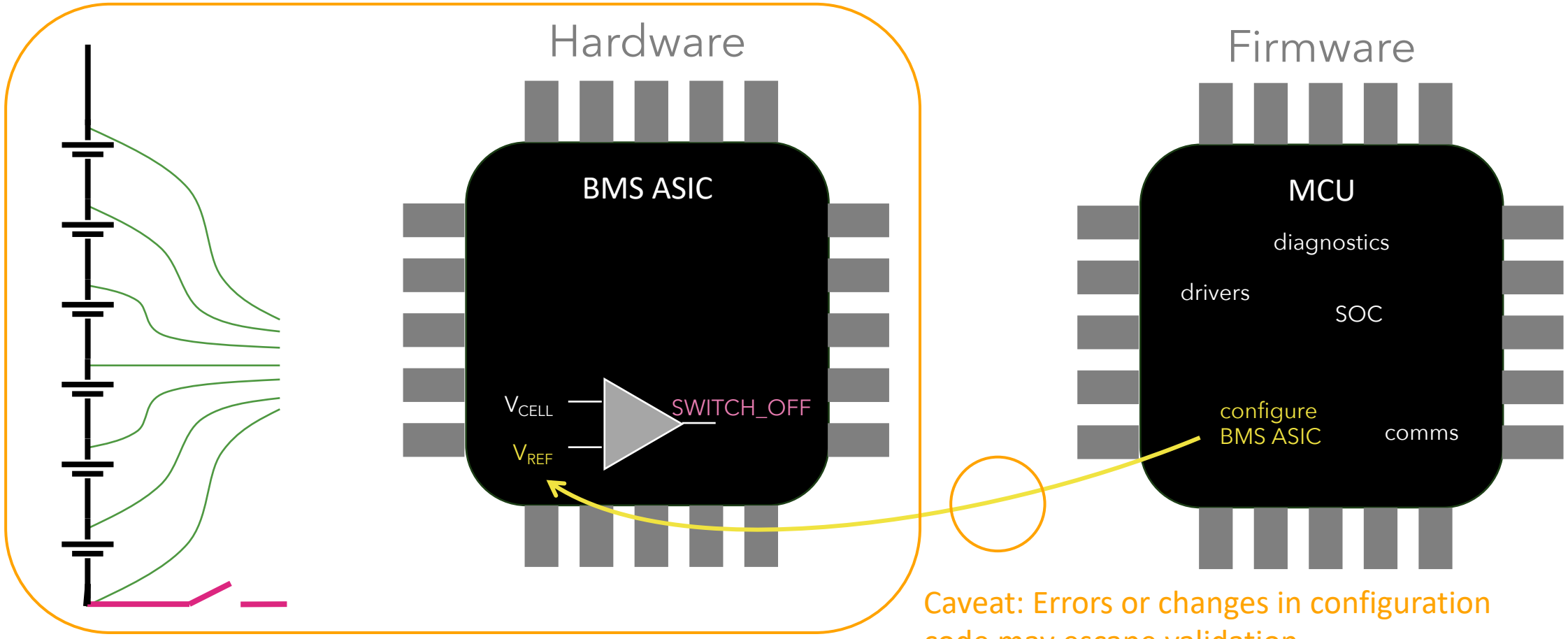
Hardware-firmware hybrid

Firmware-configurable hardware



Validate before firmware is finalized

Caveat: Errors or changes in configuration code may escape validation.

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Summary: Safety standards

Safety encompasses many systems, developed by many teams
→ Excellent safety requires excellent collaboration

ISO26262 and UL1973 are the safety standards for large automotive batteries and home energy storage, respectively

The need for safety varies by application, with automotive and home energy batteries sharing many characteristics.

Safety validation is a schedule challenge. Since we must verify the final design, learnings from validation feed into the design process quite late.

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BMS Safety

Video 1
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  - Industry standarc
  - Key consideratio

Video 2
- Fault tolerance
  - Fault tolerance le
  - Types of faults

Video 3
- Analysis tools
  - FTA = Fault Tree /

Video 4
  - DFMEA = Design

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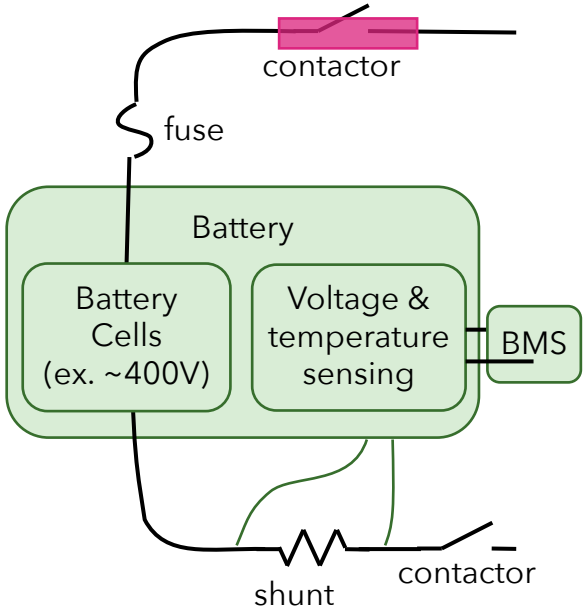
Fault tolerance levels

No fault tolerance

Single-point fault **detection**



1) Can **detect** any single fault
2) Maybe not full function with fault
3) Maybe unable to detect 2 simultaneous faults

Single-point fault **tolerance**

1) Can **operate** with single fault

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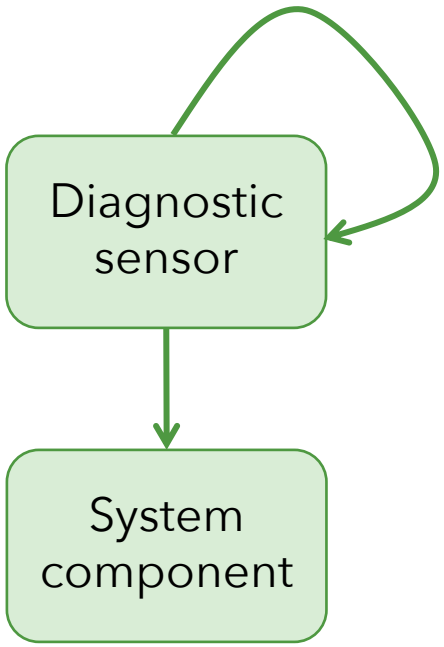
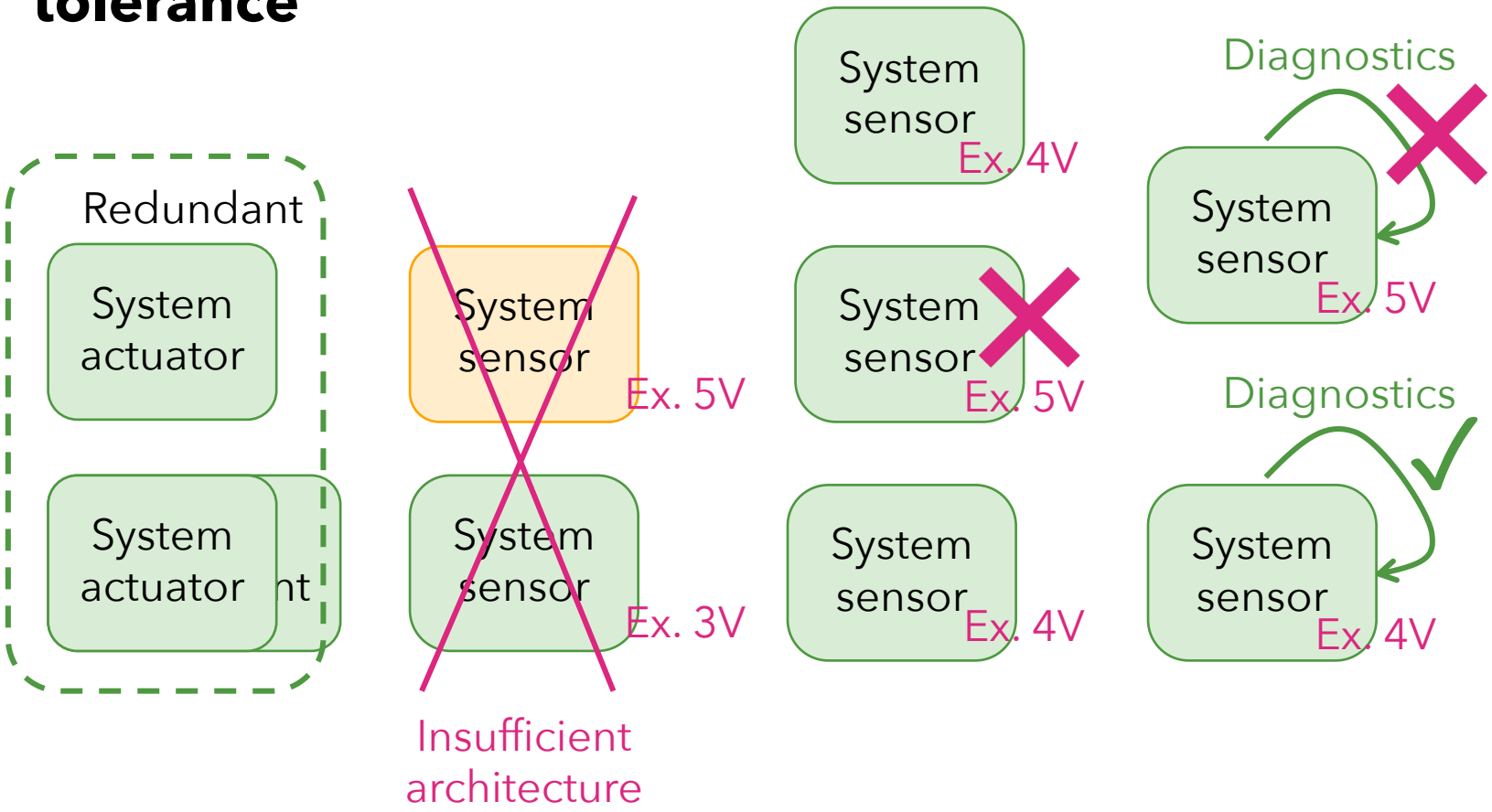
Fault tolerance implementation

Single-point fault **detection**



Single-point fault **tolerance**



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Fault tolerance levels for batteries (examples)

No fault tolerance	Single-point fault **detection**	Single-point fault **tolerance**
Toy drone Alarm clock	Electric car Home backup battery	Satellite Self-driving car control systems
Loss of function is minor inconvenience, battery is small	Large battery fire poses safety concern.	Loss of operation is a critical problem.

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BMS Safety

Video 1
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Video 2
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…intuition for…

Video 3
- Analysis tools
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Video 4
  - DFMEA = Design Failure Mode and Effects Analysis

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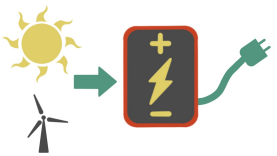
Faults are not equal: Examples



Contactor won't close

- Stops operation
- Not unsafe in itself

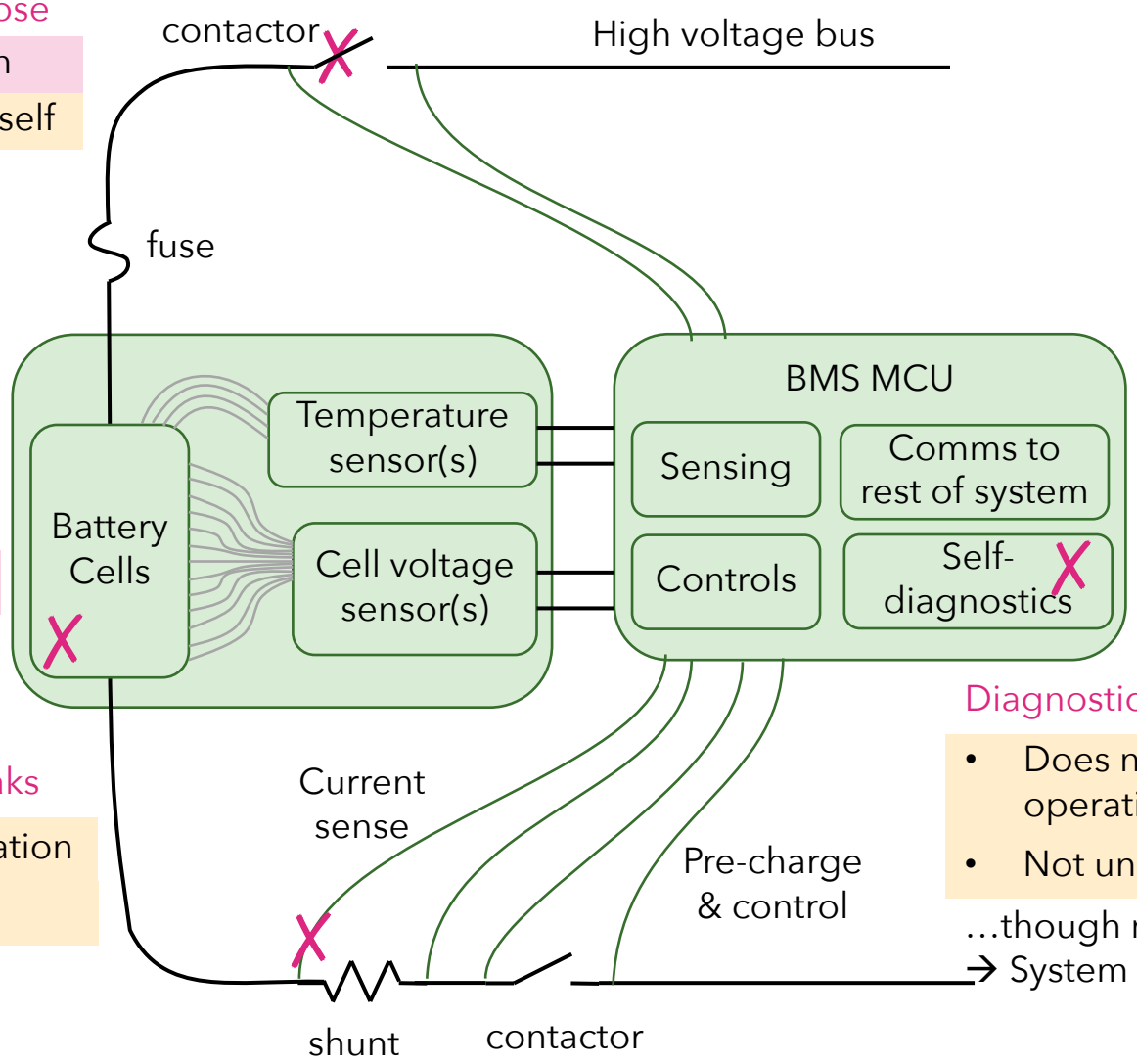
Cell internal short

- Stops operation
- Unsafe

Current sense wire breaks

- Does not stop operation
- Not unsafe in itself

→ But undermines important controls



Diagnostics don't run

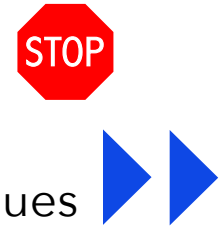
- Does not stop operation
- Not unsafe in itself

…though not inconsequential
→ System is running blind

Categories include

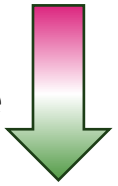
What breaks:

- Core component
  - ➢ Stops operation
- Diagnostic/sensor
  - ➢ Operation continues



Consequences

- Immediately unsafe
- Starts chain of trouble
- No or minimal impact



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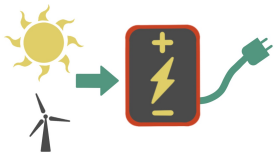
Faults types: DOES or does NOT stop system



Fault → Operation continues

Can be good:

- Safety design can guide how long to operate with fault present
  (ex. Broken sense wire → engineer can decide how long it's safe to run blind)

Can be bad:

- System continues to operate with failure
  (ex. Undetected fault in temperature sensor → may allow undesirable temperatures causing cell damage)

Fault → STOP Operation stops

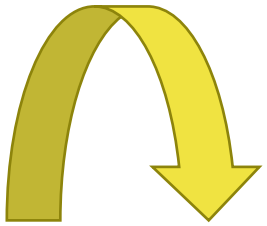
Can be good:

- Don't operate with fault
  (ex. Loss of contactor control → cannot shut contactor)

Can be bad:

- Uncontrolled
  (ex. Car immediately loses power without regard to location)

Sensors or diagnostics



Core component

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Faults types: Time until trouble

Functional safety defines: FTTI = Fault Tolerant Time

- Dangerous
  - Ex. Exothermic behavior
- Warrant immediately stopping operation, but are not in themselves dangerous
  - Ex. Current sensor failure → FTTI ≈ 1ms
- Warrant stopping operation eventually, but not necessarily immediately
  - Ex. Loss of temperature sensing → FTTI in minutes
  - Ex. Charger failure while driving → FTTI …? Discharge is ok indefinitely. Charge is not ok immediately.
- Allow continued operation
  - Ex. Failure of 1 actuator in a system with a second fully redundant sensor

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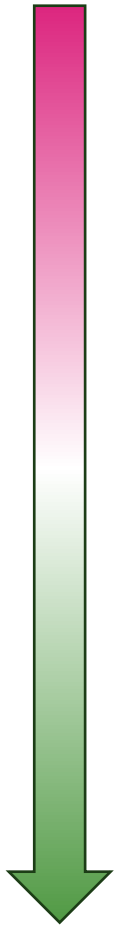
Faults types: Severity



Affects safe operation without warning

Affects safe operation with warning

Makes product inoperable

Causes decreased performance

Causes discomfort for most customers

Results in dissatisfaction for a minority of customers

No effect

DFMEA quantifies severity

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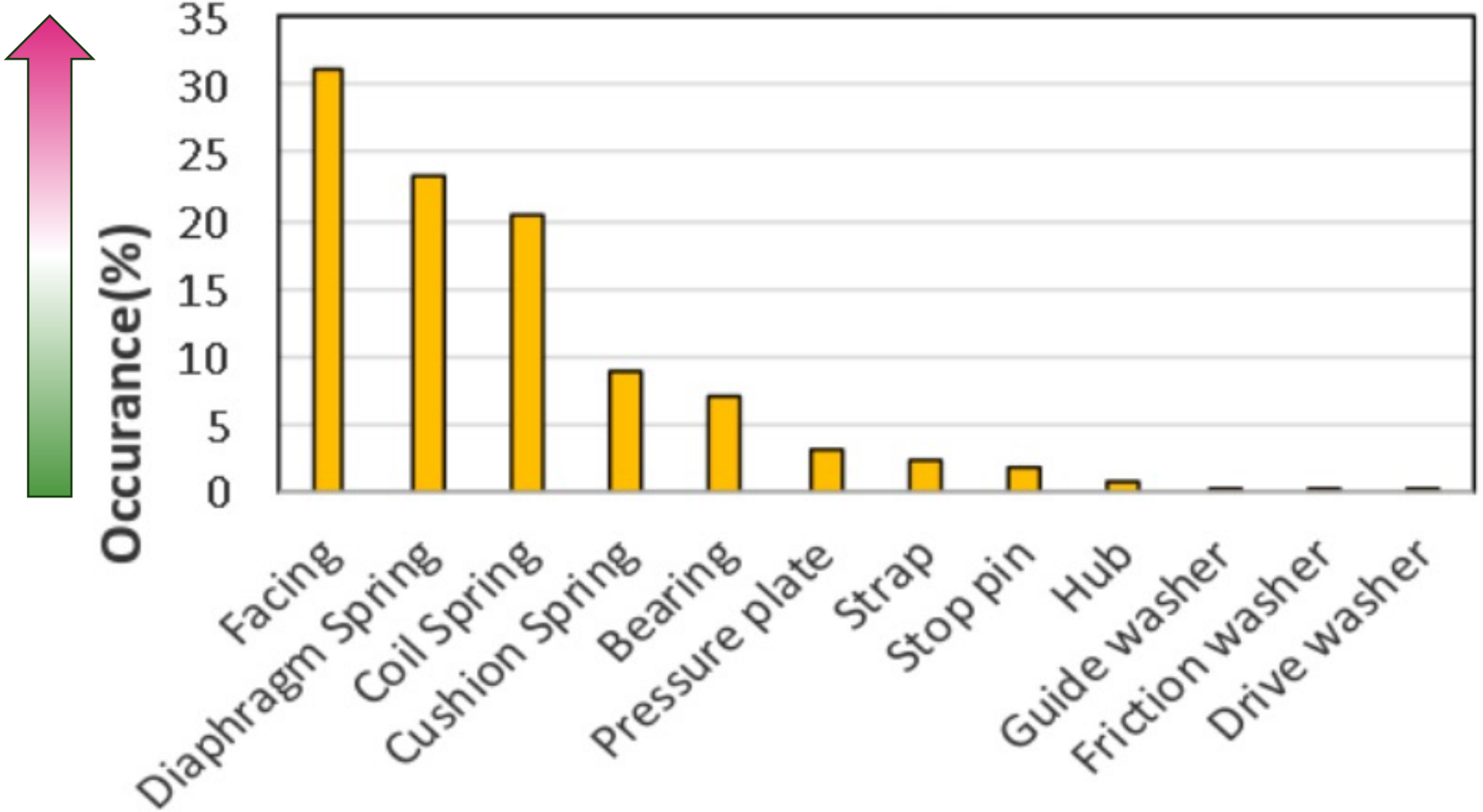
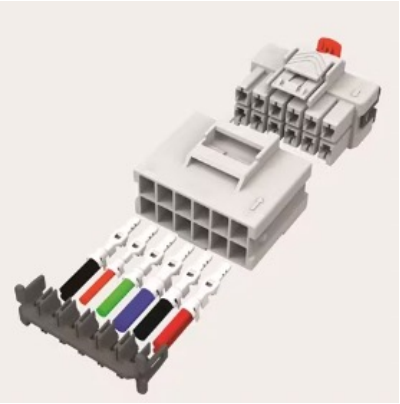
Faults types: Probability of occurrence

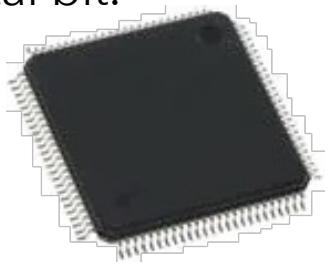
Failure of automotive clutch components



From: https://doi.org/10.2478/mme-2018-0065



Ex. Connector failures are more common than a comic ray flipping a digital bit.



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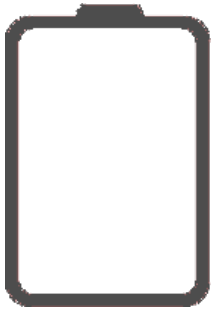
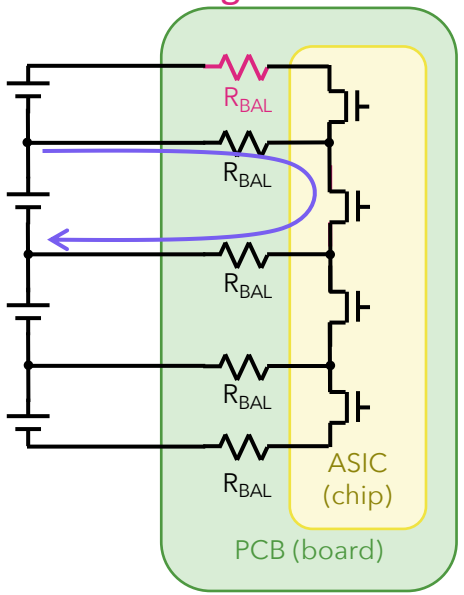
Faults types: Detectability

V_{CELL}=4.25V

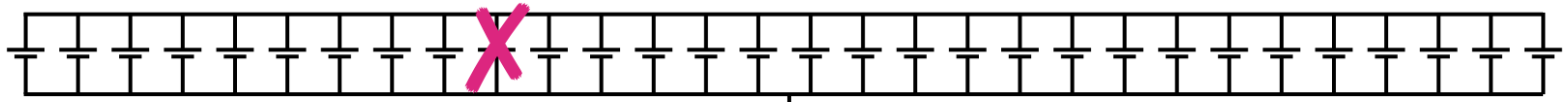


Drift in value of balancing resistor



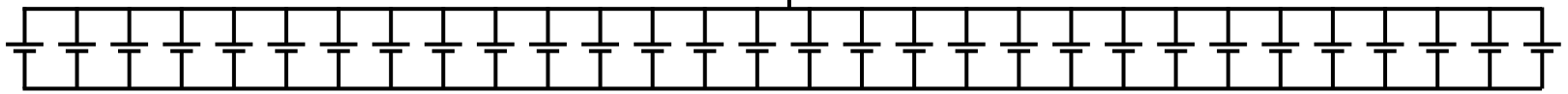
30p -> 29p

V_{CELL}=3.70V



Q=200Ah

V_{CELL}=3.70V



Q=193Ah

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Fault types: Summary

Important to consider **type** of fault when deciding about mitigations

- **What** broke: System itself or diagnostics
- **Consequences:**
  - **Severity**
  - **Time** scale
- **Probability** of occurrence
- **Detectability**

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BMS Safety

Video 1
- Car/home battery sa
  - Industry standarc
  - Key consideratio

Video 2
- Fault tolerance
  - Fault tolerance le
  - Types of faults

Video 3
- Analysis tools
  - FTA = Fault Tree /

Video 4
  - DFMEA = Design

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BMS Safety

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Video 4
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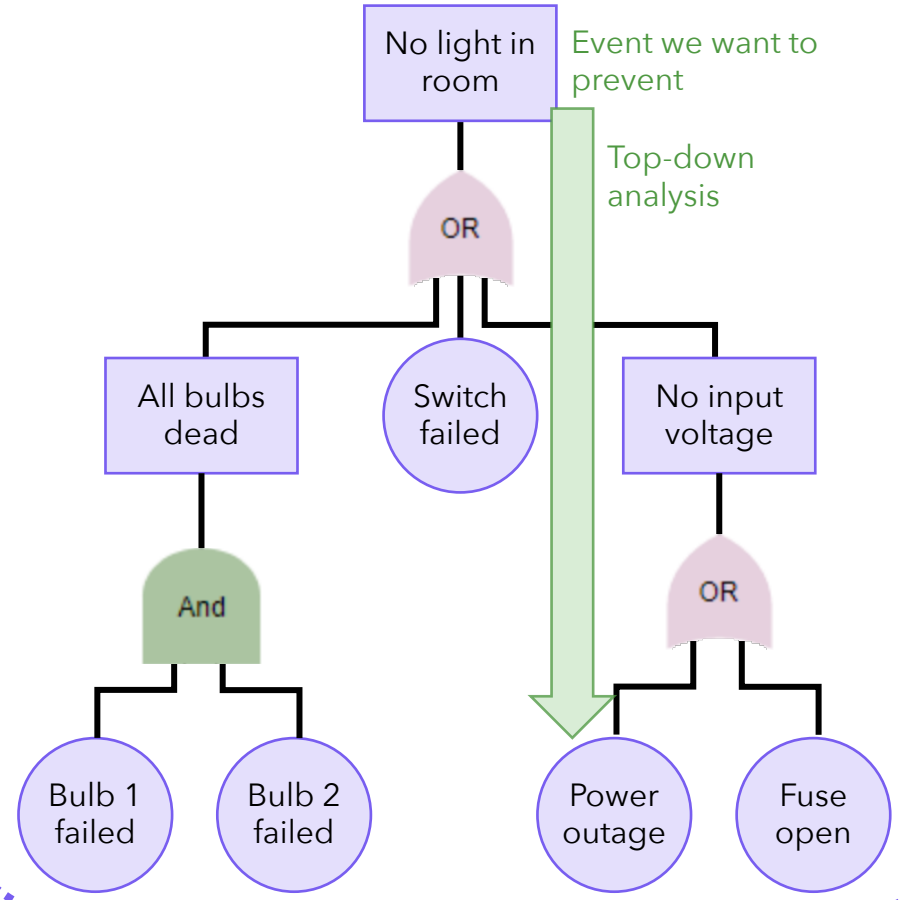
Safety analysis tools

Fault Tree Analysis (FTA)



Engineering intuition and judgement

Failure Modes and Effects Analysis (FMEA)

Severity

Occurrence

Detection rating

Function	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D

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BMS Safety

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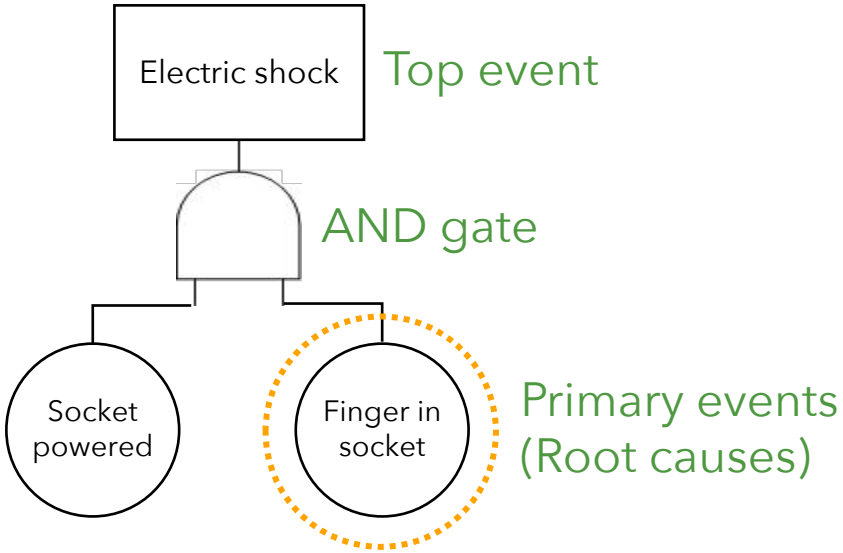
Fault Tree Analysis (FTA): Start at the top

Identify the top event (the failure we're trying to prevent)

…and build the tree down to the root causes (basic events)

Top-down analysis



Process mitigation: Training
Design mitigation: Redesign socket with smaller holes

What's the goal?

→ Shows us where mitigations are needed

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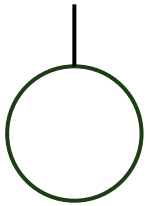
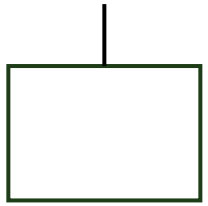
FTA symbols: Events



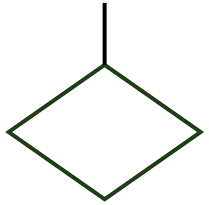
Basic or primary event
(occur without dependencies)

Ex. Sense wire breaks



- Top event.
- Intermediate event/state.

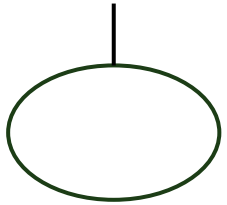
Ex. Cell over-voltage



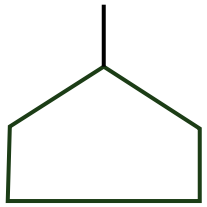
Undeveloped event.
(an incompletely explored branch)

Ex. Voltage sensor failure

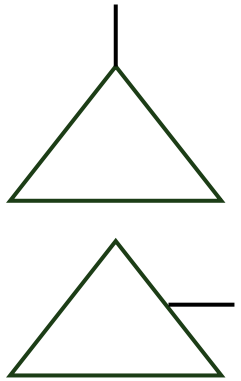
Enough to do a decent analysis



Conditional event
(associated with another event)



External event.
Or represents either occurrence or non-occurrence of an event.



Transfer-in and transfer-out
(replicate a branch of FTA, for example across multiple pages)

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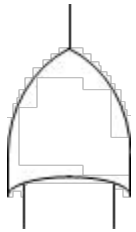
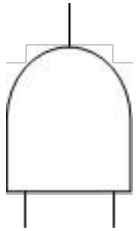
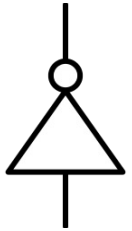
FTA symbols: Gates



OR gate: Output event occurs when **any** input events occurs

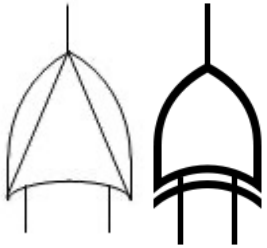


AND gate: Output event occurs only if **all** input events occur

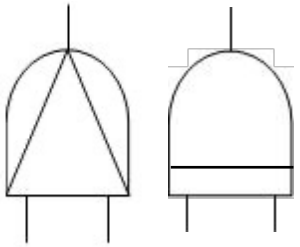


NOT gate: Output event occurs if the input does **not** occur

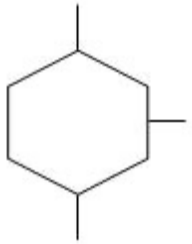
Enough to do a decent analysis



Exclusive OR: Output event occurs if exactly one input occurs



Priority AND: The output occurs if the inputs occur in a specific sequence



Inhibit: The output occurs if the input event occurs and an enabling condition is satisfied

… and others ….

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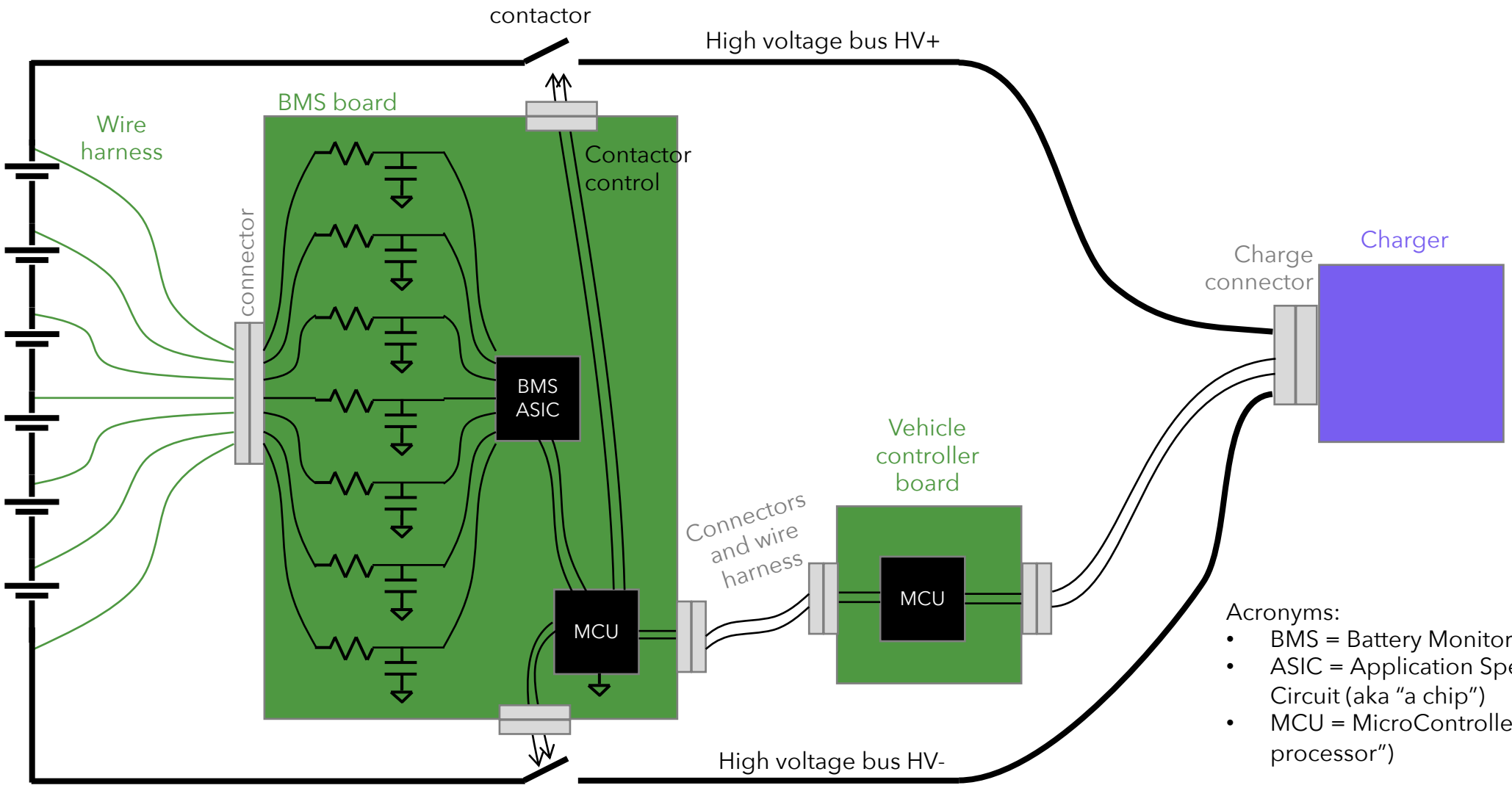
Before analysis: A system diagram

(simplified relative to a production system)



Acronyms:

- BMS = Battery Monitoring System
- ASIC = Application Specific Integrated Circuit (aka "a chip")
- MCU = MicroController Unit (aka "a processor")

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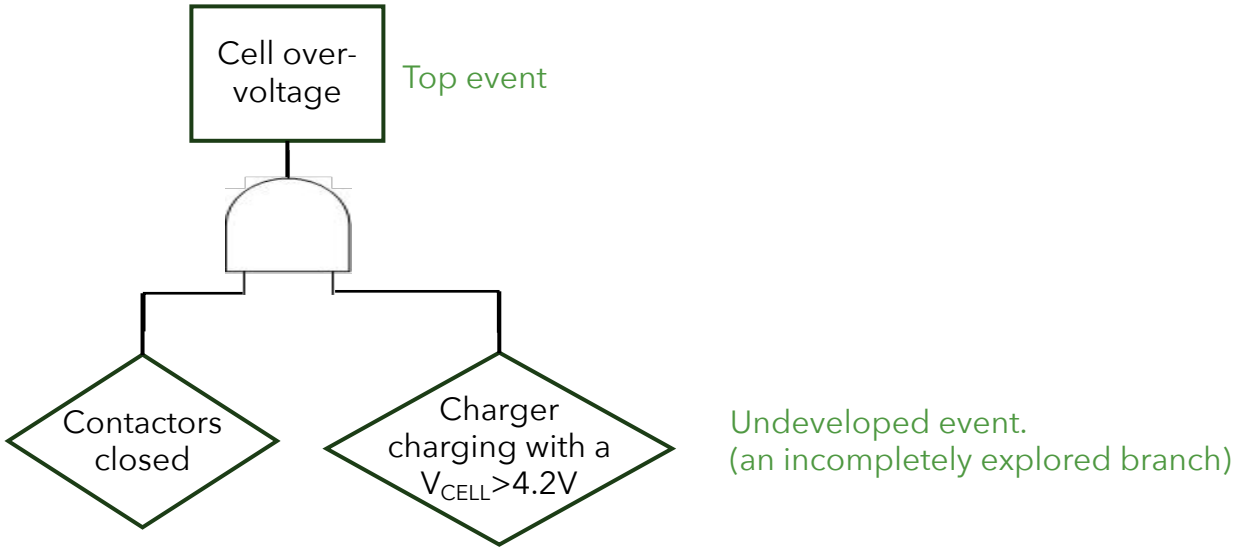
FTA example



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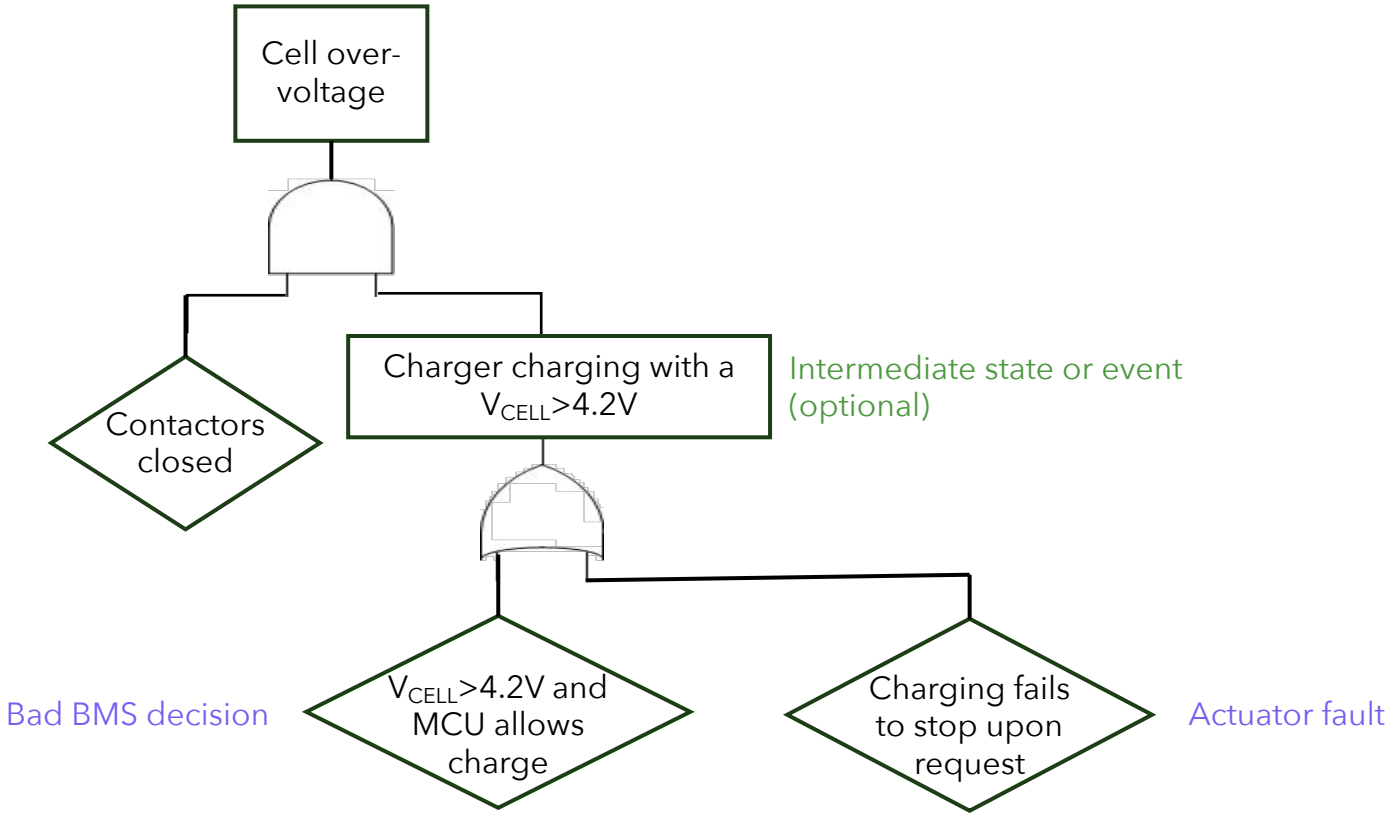
FTA example



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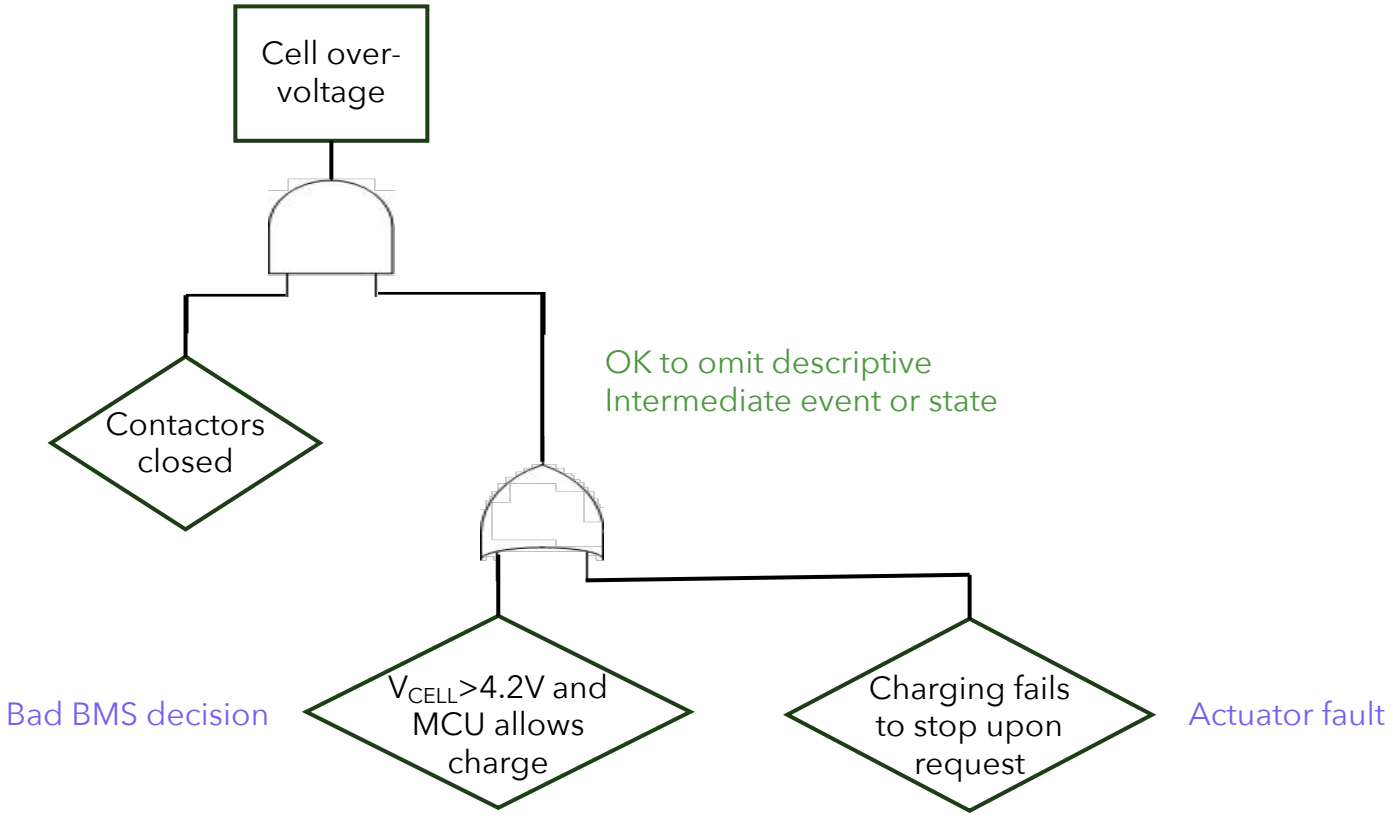
FTA example



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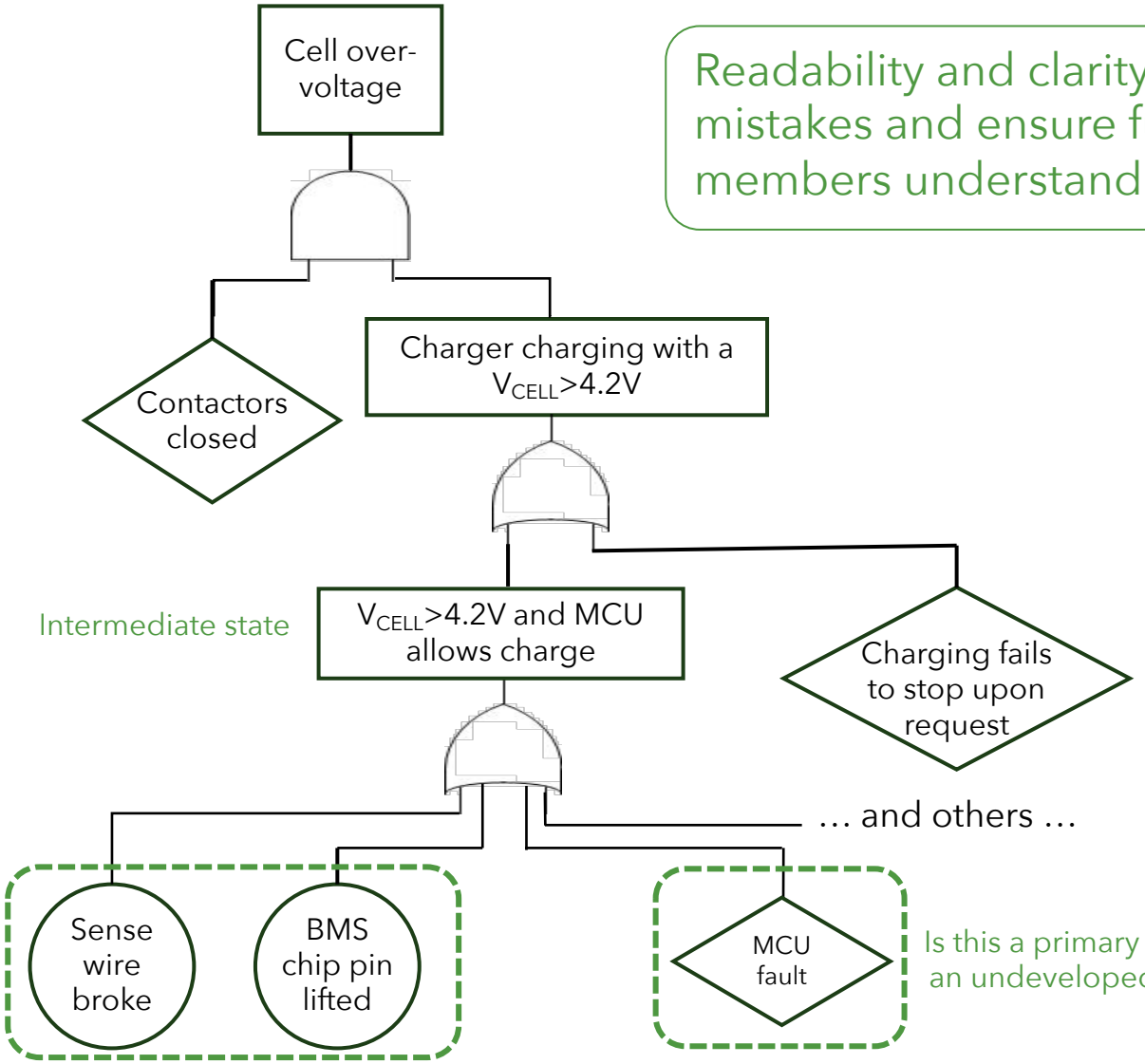
FTA example

Readability and clarity reduce mistakes and ensure future team members understand past decisions.



Similar. Both are an **open sense connection** between the voltage sensor (BMS chip) and the cell.

Is this a primary event or an undeveloped event?

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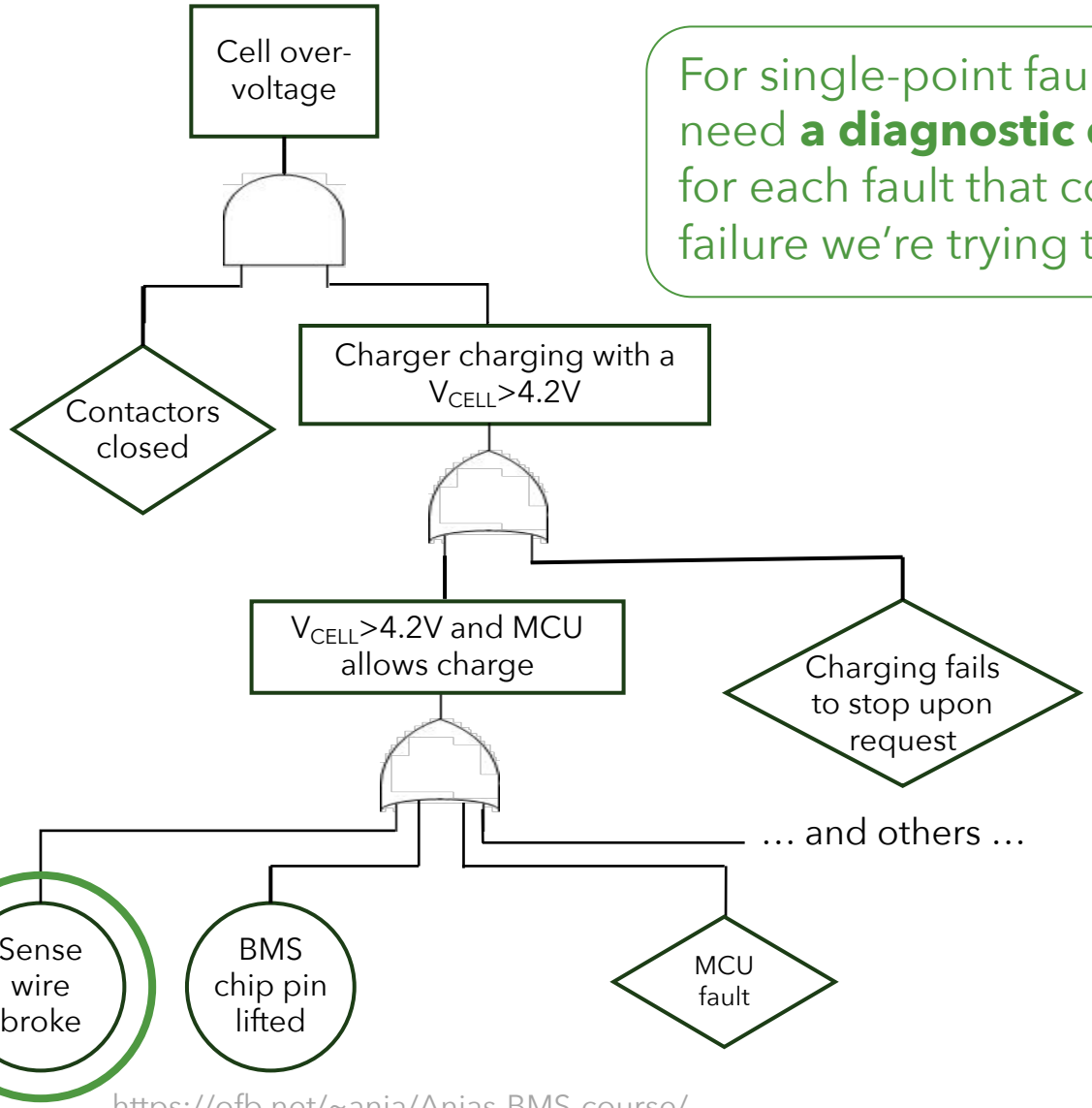
FTA example: Use it!

For single-point fault **detection**, need **a diagnostic or mitigation** for each fault that could lead to the failure we're trying to prevent



Diagnostic #1: Detect broken sense wire

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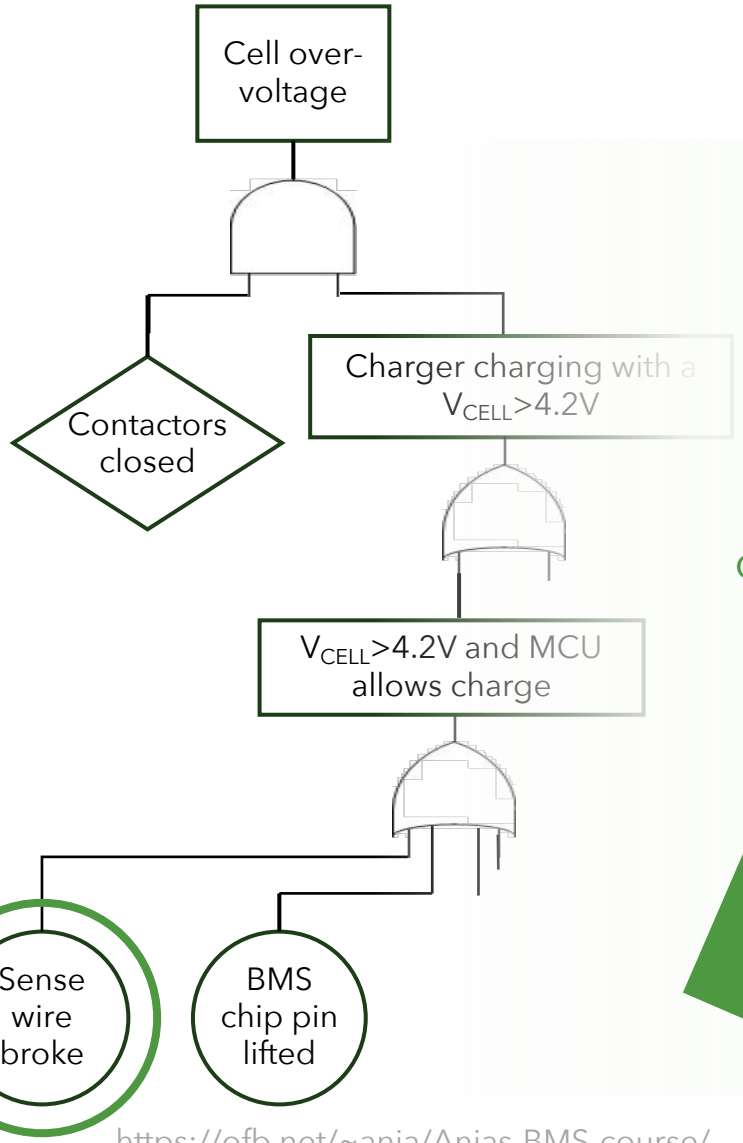
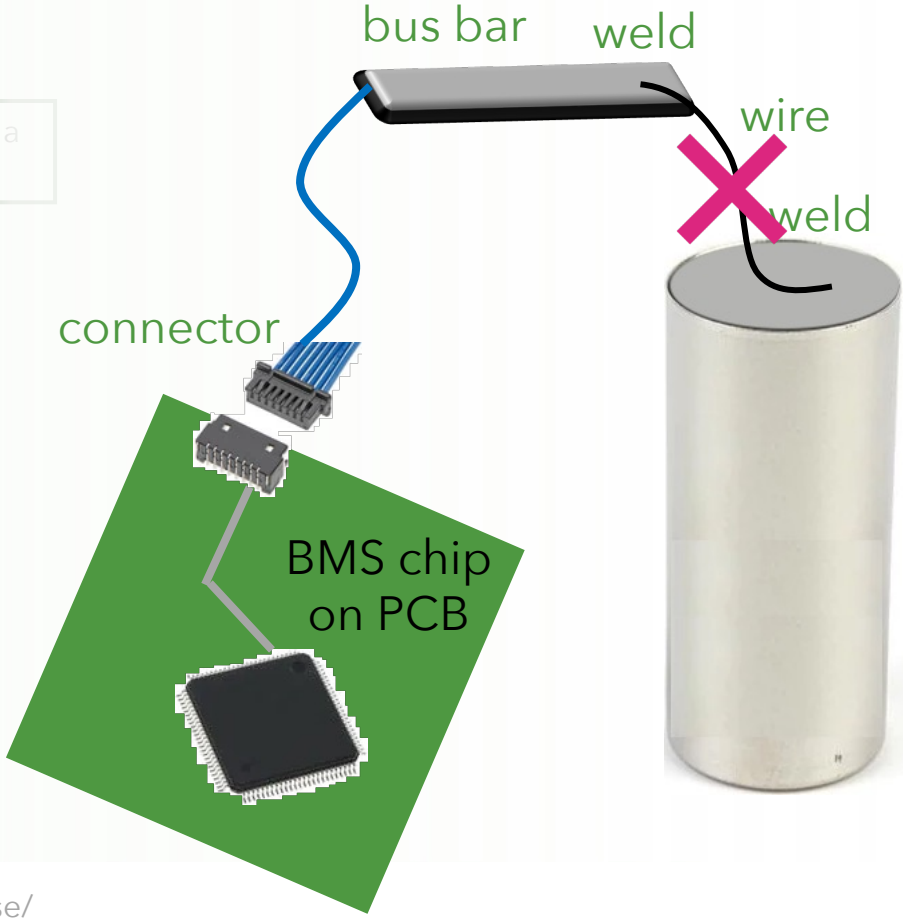
FTA Diagnostic #1: Broken wire detection



Diagnostic #1: Detect broken sense wire



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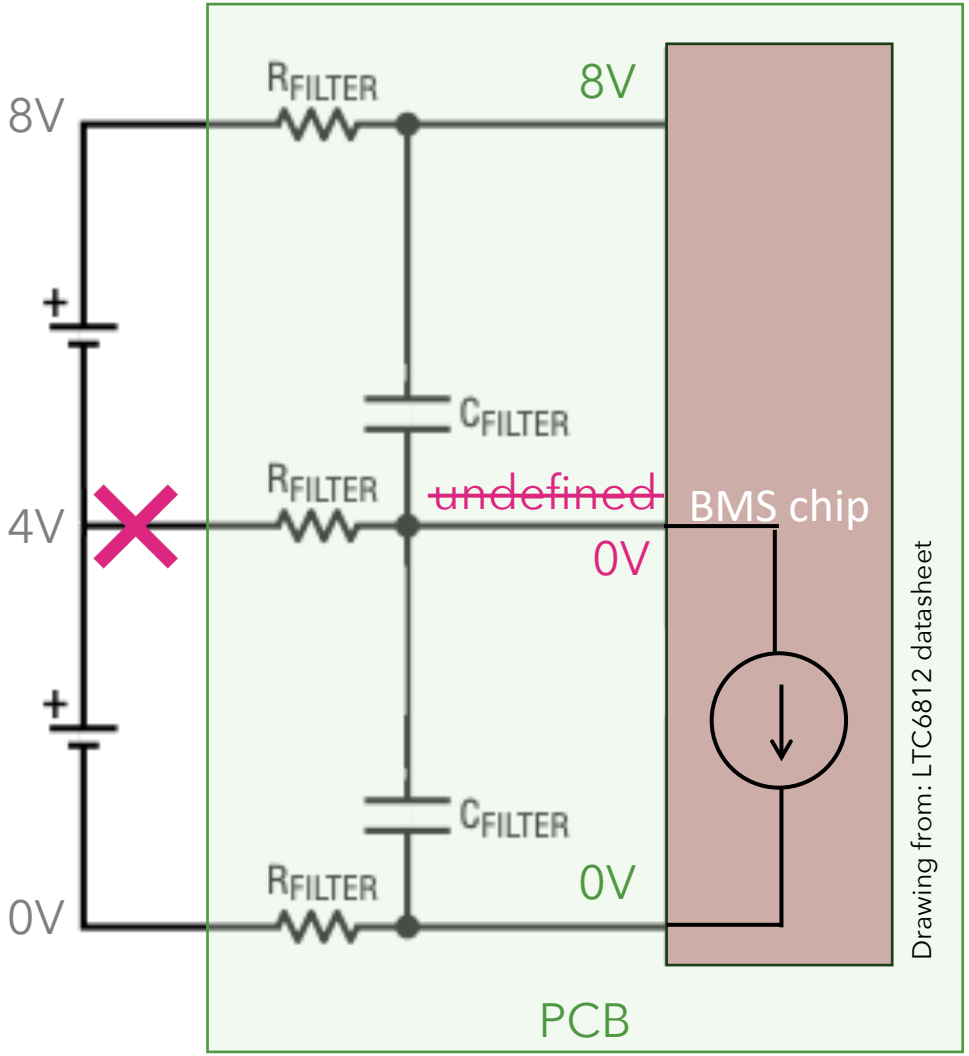
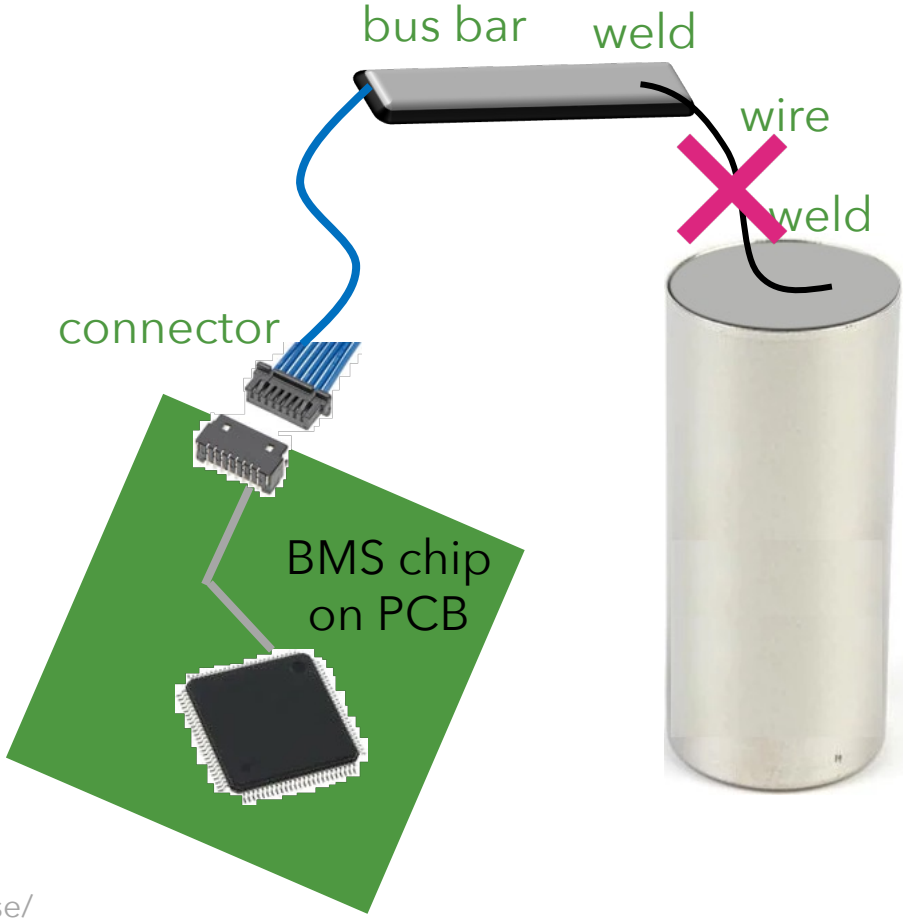
https://ofb.net/~ania/Anias-BMS-course/

Broken sense wire diagnostic





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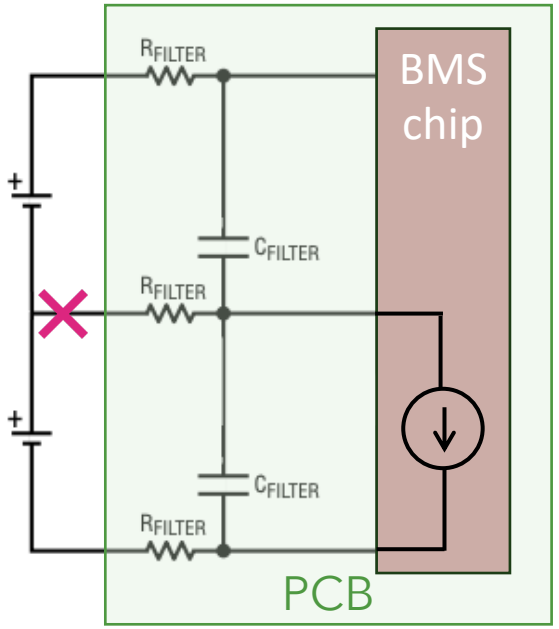
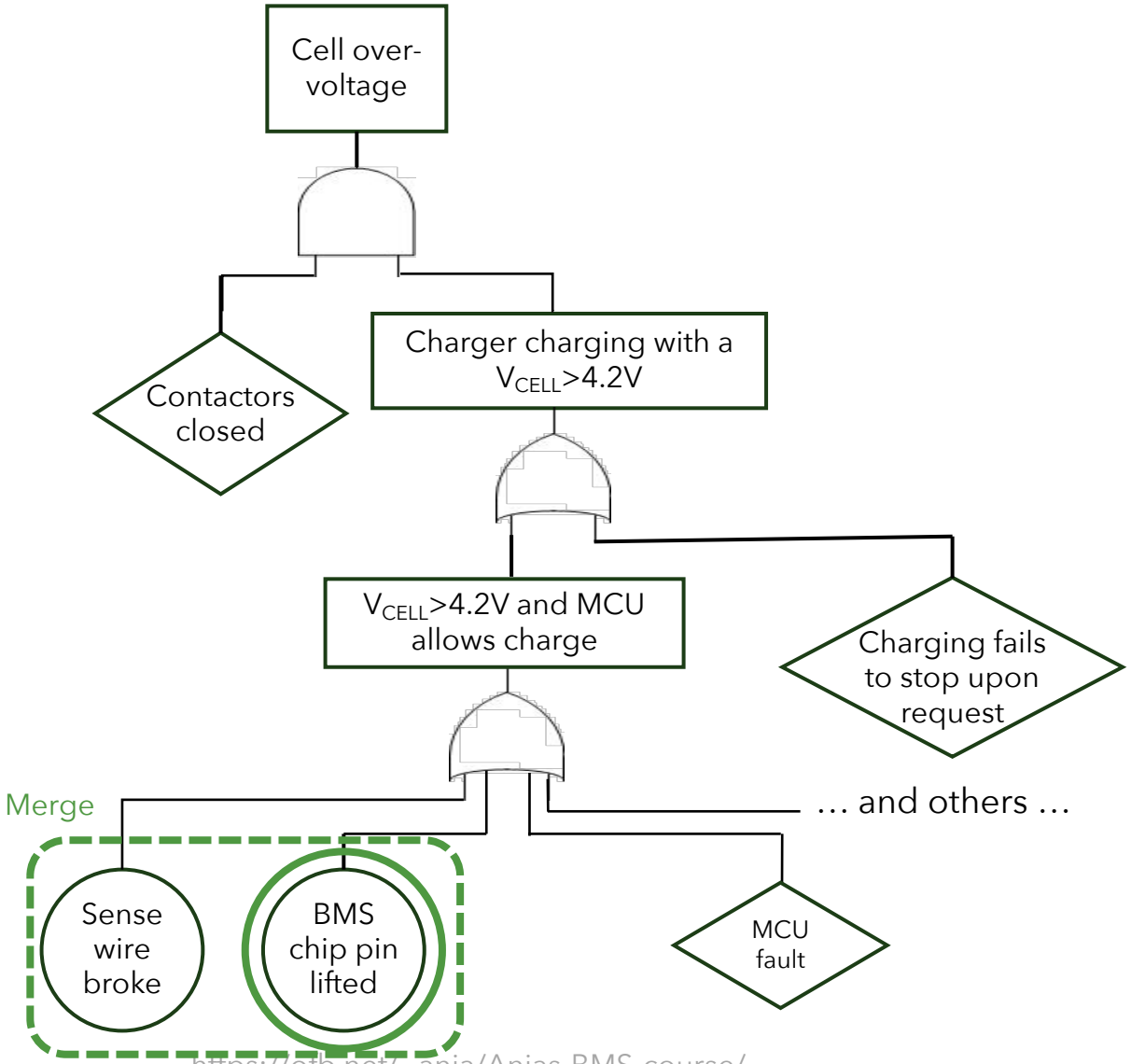
FTA: Diagnostic #2 BMS chip pin lifted





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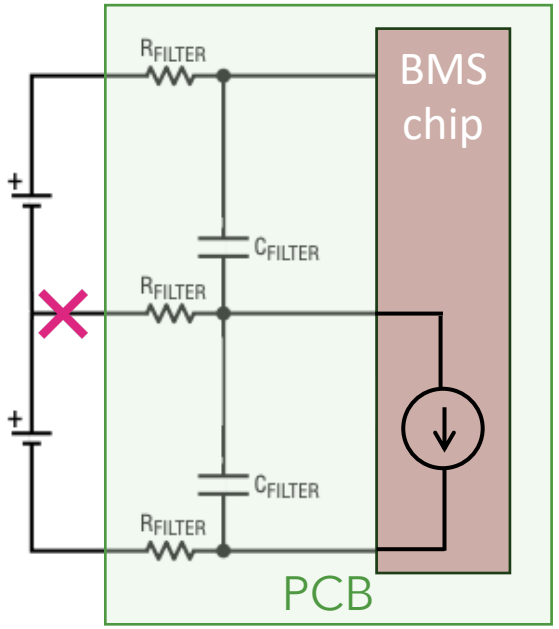
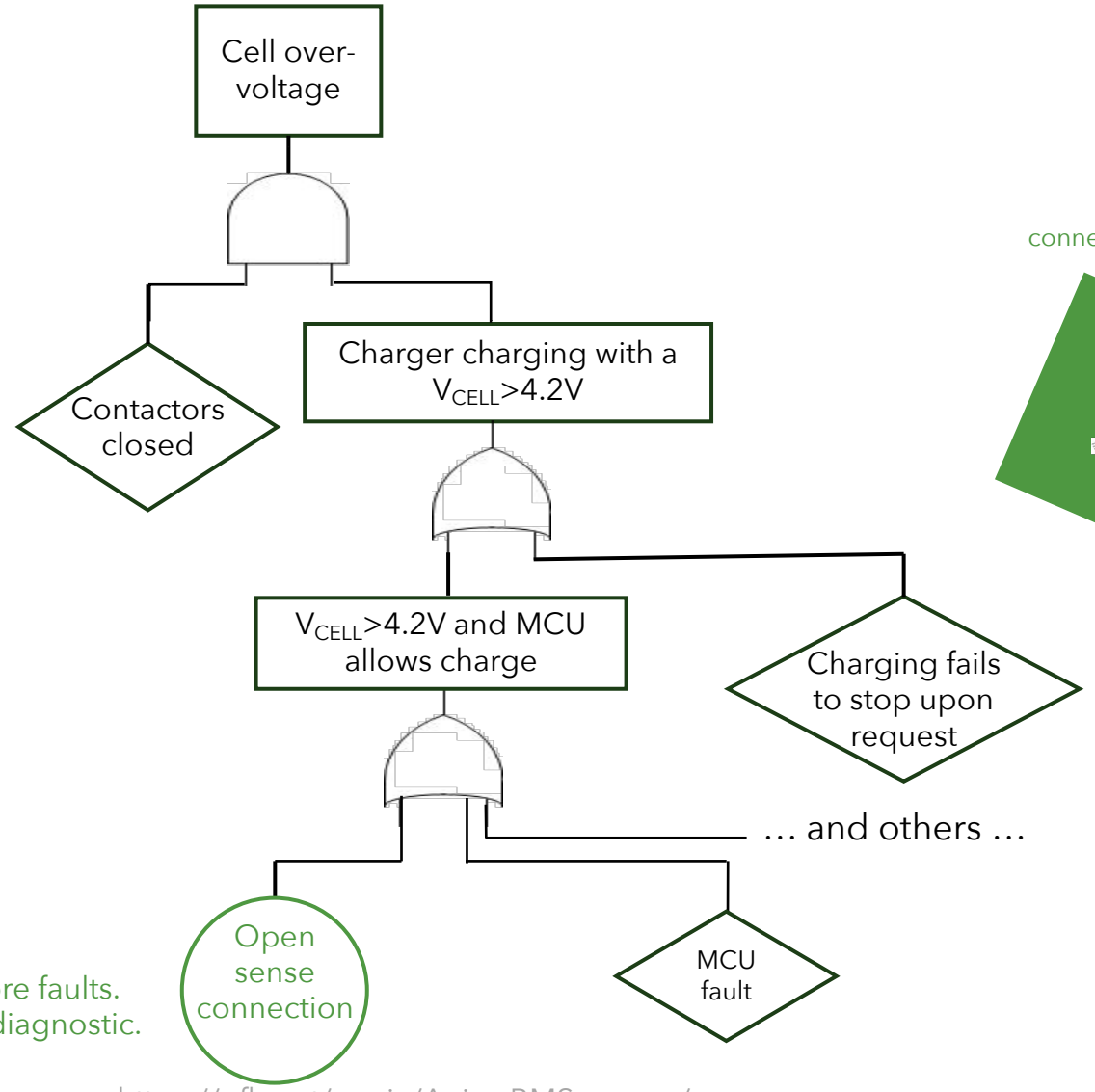
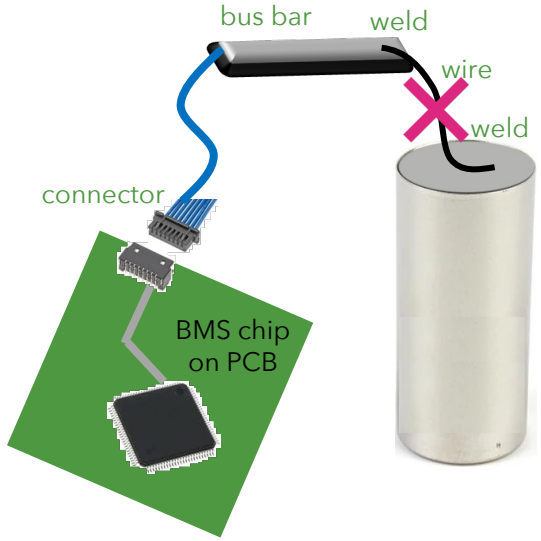
FTA: Update to merge faults







Simpler.
Improved: Captures more faults.
Corresponds to single diagnostic.

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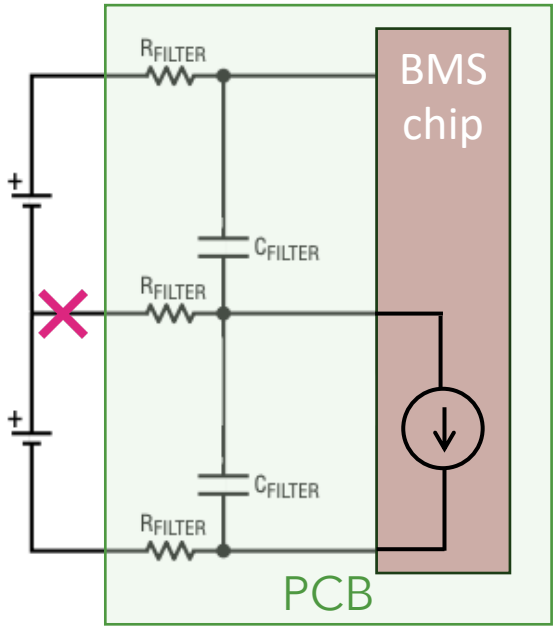
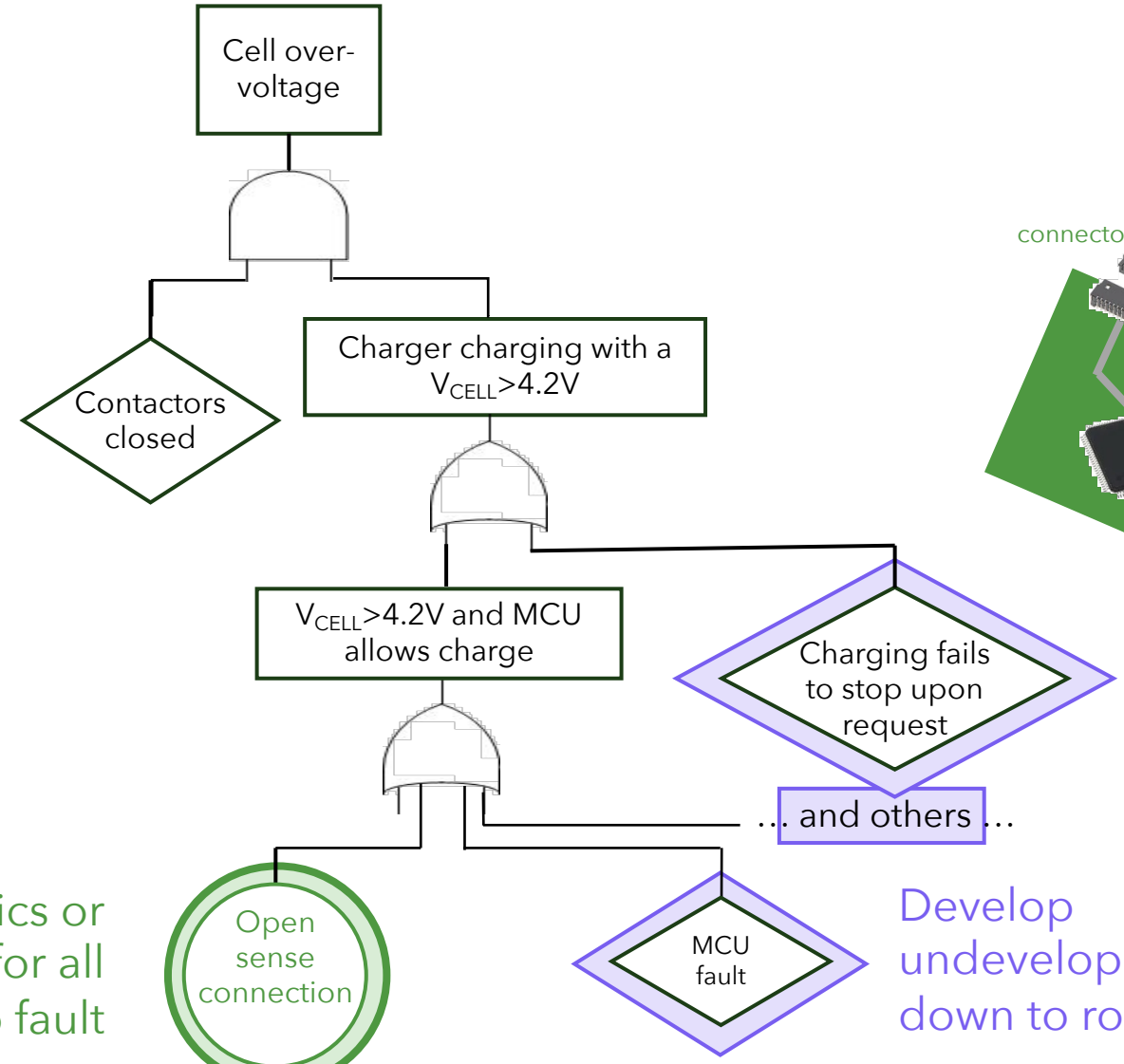
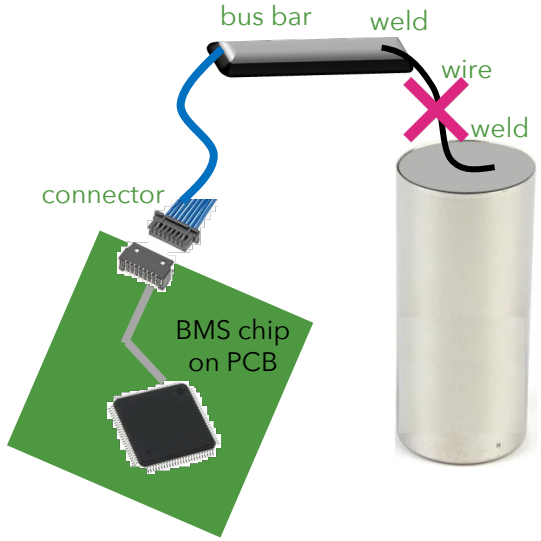
FTA: Finish building & using the fault tree







Ensure diagnostics or mitigations for all paths to top fault

Develop undeveloped events down to root causes

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https://olb.net/~ania/Anias-BMS-course/

FTA (Fault Tree Analysis) Summary

1) Top-down tool
   → Start with some concerning event at top of diagram
   → Find root cause events (single-point faults)

2) Facilitates development of mitigations to all relevant faults
   → Good tool to guide design

3) Requires good understanding of system
   → Best done by interdisciplinary team

4) Does not reflect severity nor probability of events

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  - Key consideratio

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  - Fault tolerance le
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  - Fault tolerance levels
  - Types of faults

Video 3
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  - FTA = Fault Tree Analysis

Video 4
  - DFMEA = Design Failure Mode and Effects Analysis

FTA omits severity & probability of events

...and produces a readable diagram

DFMEA includes severity & probability of events

...and produces a dense table with thorough coverage

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FMEA or DFMEA?

DFMEA = Design Failure Mode and Effect Analysis

FMEAs include:

- **DFMEA = Design FMEA**
- PFEMEA = Process FMEA
- Manufacturing FMEAs
- Software FMEAs
- System / Functional FMEAs
- Service FMEAs

Subtypes of DFMEAs or PFMEAs

List from: https://relyence.com/2018/08/28/overview-fmea-types/

Failure Modes and Effects Analysis (FMEA)

Severity

Occurrence

Detection rating

Function	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D

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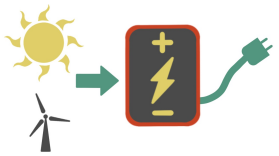
Battery Switch DFMEA: Function



List functions

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens										
Closes										
Pre-charges										

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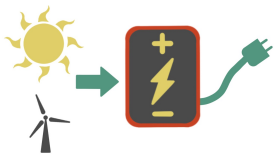
Battery Switch DFMEA: Requirements



List requirements

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V									
	Opens within 70 ms of command									
	Can open 10,000 times before failure									

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Battery switch DFMEA: Failure modes

List failure modes

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut								
		Control signal failure								
		Insufficient clearances; arcs at <500V								

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Battery switch DFMEA: Effects of failure

List effects of failures

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open							

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Battery switch DFMEA: Severity

Assign severity

Function	Requirements	Failure mode	Effects of failure	S
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	

"**Severity**" is a ranking number associated with the most serious effect for a given failure mode, based on the criteria from a severity scale. It is a relative ranking within the scope of the FMEA and is determined without regard to the likelihood of occurrence or detection.

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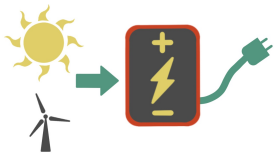
FMEA: Severity (example scale)



Ranking	Effect	Design FMEA Severity	Process FMEA Severity
10	Hazardous-no warning	affects safe operation without warning	may endanger machine or operator without warning
9	Hazardous- w/ warning	affects safe operation with warning	may endanger machine or operator with warning
8	Very High	makes product inoperable	major disruption in operations (100% scrap)
7	High	makes product operable at reduced performance (customer dissatisfaction)	minor disruption in operations (may require sorting and some scrap)
6	Moderate	results in customer discomfort	minor disruption in operations (no sorting but some scrap)
5	Low	results in comfort and convenience at a reduced level	minor disruption in operations (portion may require rework)
4	Very Low ?	results in dissatisfication by most customers.	minor disruption in operations (some sorting and portion may require rework)
3	Minor	results in dissatisfication by average customer.	minor disruption (some rework but little affect on production rate)
2	Very Minor ?	results in dissatisfication by few customers.	minor disruption (minimal affect on production rate)
1	None	No effect	No effect

Table from: https://www.leansixsigmadefinition.com/glossary/fmea/

Effects of failure	S
Battery switch cannot open	10

Normal operation:
→ Not much impact — 2

Very low battery:
→ Cannot prevent over-discharge — 2 or 4?

After accident:
→ Safety risk for first responders — 10

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Battery switch DFMEA: Severity

Assign severity

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10						

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Battery switch DFMEA: Causes of failure

List cause(s) of failure

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut					

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Battery switch DFMEA: Occurrence

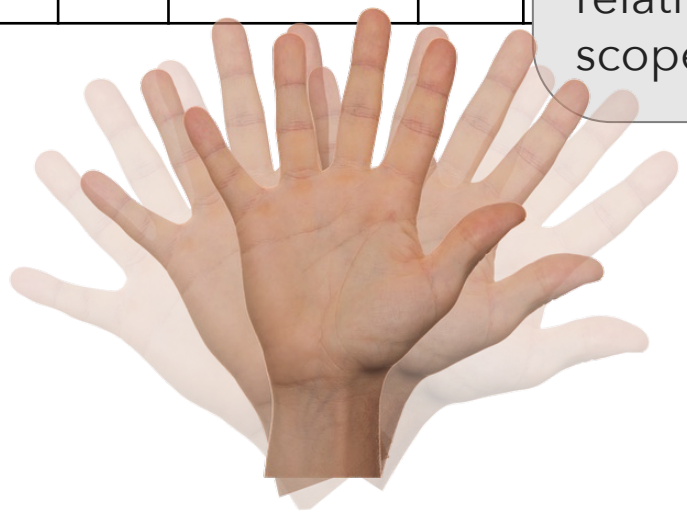
Design → Production → Field failures

(aka: Occurrence)

Assign occurrence

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	

"**Occurrence**" is a ranking number associated with a cause of failure, based on the criteria from an occurrence scale. It is a relative ranking within the scope of the FMEA.



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FMEA: Occurrence (example scale)

Ranking	Effect	Failure Rates
10	Extremely High	> 1 in 2
9	Very High	1 in 3
8	Very High	1 in 8
7	High	1 in 20
6	Marginal	1 in 100
5	Marginal	1 in 400
4	Unlikely	1 in 2000
3	Low	1 in 15,000
2	Very Low	1 in 150,000
1	Remote	< 1 in 1,500,000

Causes of failure	O
Arcing welds contactor shut	**2**

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Battery switch DFMEA: Process controls

Describe detection mechanism

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None			

"**Process controls**" are tests or analyses that detect a failure mode or associated cause.

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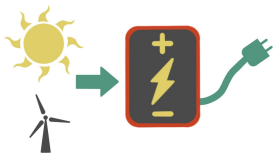
Battery switch DFMEA: Detection



Assign detection rating

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None			

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FMEA: Detection (example scale)

Ranking	Effect	Design FMEA Detection
10	Absolute uncertainty	No chance that design control will detect cause mechanism and subsequent failure.
9	Very remote	Very remote chance that design control will detect cause mechanism and subsequent failure.
8	Remote	Remote chance that design control will detect cause mechanism and subsequent failure.
7	Very Low	Very low chance that design control will detect cause mechanism and subsequent failure.
6	Low	Low chance that design control will detect cause mechanism and subsequent failure.
5	Moderate	Moderate chance that design control will detect cause mechanism and subsequent failure.
4	Moderately High	Moderately high chance that design control will detect cause mechanism and subsequent failure.
3	High	very remote chance that design control will detect cause mechanism and subsequent failure.
2	Very High	Very high chance that design control will detect cause mechanism and subsequent failure.
1	Almost Certain	Design control will almost certainly detect cause mechanism and subsequent failure.

Table from: https://www.leansixsigmadefinition.com/glossary/fmea/

No detection mechanism

Fault will be detected within 1 day of failure in ≥95% of units

Fault will be detected within 100ms of failure in ≥95% of units

Fault will be detected within 100ms of failure in 100% of units

Fault will be detected within 10ms of failure in 100% of units

Process controls	**D**
None	**10**

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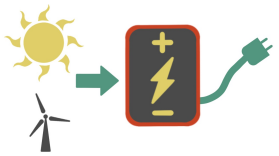
Battery switch DFMEA: Detection



Assign detection rating

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None	10		

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Battery switch DFMEA: Criticality & RPN

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None	10	20	200

Criticality =S×O

Criticality

RPN =S×O×D

Risk Priority Number (RPN)

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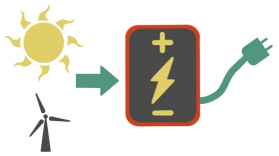
Battery switch DFMEA: Perspective



Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None	10	20	200
		Control signal failure								
		Insufficient clearances; arcs at <500V								
	Opens within 70 ms of command									
	Can open 10,000 times before failure									
Closes										
Pre-charges										

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Battery switch DFMEA: What's the point?

Criticality → Risk Priority Number (RPN)

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None	10	20	200
		Control signal failure	Battery switch cannot open or close	10	~~~	~~~	~~~	~~~	~~~	~~~
		Insufficient clearances; arcs at <500V	Battery switch damage	?	~~~	~~~	~~~	~~~	~~~	~~~
	Opens within 70 ms of command	~~~	~~~	~~~	~~~	~~~	~~~	~~~		
	Can open 10,000 times before failure	~~~	~~~	~~~	~~~	~~~	~~~	~~~		
Closes	Conducts <1Ω	~~~	~~~	~~~	~~~	~~~	~~~	~~~		
	Closes within 70ms of command	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~
Pre-charges	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~
	~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~	~~~

Ranking creates a priority list for addressing weaknesses

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FMEA: Implement mitigation

Ranking	Effect	Design FMEA Detection
10	Absolute uncertainty	No chance that design control will detect cause mechanism and subsequent failure.
9	Very remote	Very remote chance that design control will detect cause mechanism and subsequent failure.
8	Remote	Remote chance that design control will detect cause mechanism and subsequent failure.
7	Very Low	Very low chance that design control will detect cause mechanism and subsequent failure.
6	Low	Low chance that design control will detect cause mechanism and subsequent failure.
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Table from: https://www.leansixsigmadefinition.com/glossary/fmea/

No detection mechanism

Fault will be detected within 1 day of failure in ≥95% of units

Fault will be detected within 100ms of failure in ≥95% of units

Fault will be detected within 100ms of failure in 100% of units

Fault will be detected within 10ms of failure in 100% of units

Process controls	**D**
None	~~10~~ 2

Add diagnostic to detect contactor welded shut.

Runs once for every contactor open/close cycle

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Battery switch DFMEA: What's the point?

Criticality → Risk Priority Number (RPN)

Function	Requirements	Failure mode	Effects of failure	S	Causes of failure	O	Process controls	D	S×O	S×O×D
Opens	Blocks current upon request, up to 500V	Welded shut	Battery switch cannot open	10	Arcing welds contactor shut	2	None	~~10~~ 2	20	~~200~~ 40
		Control signal failure	Battery switch cannot open or close	10	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
		Insufficient clearances; arcs at <500V	Battery switch damage	?	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
	Opens within 70 ms of command	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
	Can open 10,000 times before failure	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
Closes	Conducts <1Ω	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
	Closes within 70ms of command	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
Pre-charges	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
	~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~

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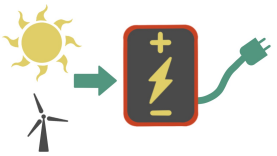
DFMEA (Design Failure Mode and Effect Analysis) Summary



1) Detailed quantitative analysis
   → Start with requirements
   → Find potential failures of each
   → Assign quantitative ratings to possible consequences
2) Facilitates thoroughness
3) Requires good understanding of system
   → Best done by interdisciplinary team
   → Usually done by dedicated safety engineer
4) Reflects severity and probability of events ← unlike FTA
5) Most useful for:
   → Checking nothing was missed
   → Prioritizing by likelihood (occurrence) and severity

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https://ofb.net/~ania/Anias-BMS-course/

"The End" by Gerd Altmann from Pixabay